

MTH 131: Exam 4

Name _____

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Use the given conditions to write an equation for the line in the indicated form (6 points each).

- 1) Passing through (4, -4) and parallel to the line whose equation is $y = -2x + 5$; slope-intercept form 1) _____

Use the given conditions to write an equation for the line in the indicated form.

- 2) Passing through (2, 5) and perpendicular to the line whose equation is $y = \frac{1}{6}x + 8$; slope-intercept form 2) _____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

Determine whether the function is a polynomial function. If you answer "No", explain why.

- 3) $f(x) = 2 - \frac{1}{x^4}$ 3) _____
A) Yes B) No

- 4) $f(x) = \pi x^3 - 4x^2 - 5$ 4) _____
A) Yes B) No

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Find the domain and range of the quadratic function whose graph is described.

- 5) The vertex is (-1, -6) and the graph opens up. 5) _____

Find the coordinates of the vertex for the parabola defined by the given quadratic function.

6) $f(x) = (x - 4)^2 - 4$

6) _____

Using your answer from problem 6, find the axis of symmetry of the parabola defined by the given quadratic function.

7) $f(x) = (x - 4)^2 - 4$

7) _____

Using your answer from problem 6, find the range of the quadratic function.

8) $f(x) = (x - 4)^2 - 4$

8) _____

Find the x-intercepts (if any) for the graph of the quadratic function.

9) $f(x) = (x - 4)^2 - 4$

9) _____

Find the y-intercept for the graph of the quadratic function.

10) $f(x) = (x - 4)^2 - 4$

10) _____

Find the coordinates of the vertex for the parabola defined by the given quadratic function.

11) $f(x) = 4 + 4x + x^2$

11) _____

Using your answer from problem 11, determine whether the given quadratic function has a minimum value or maximum value. Then find the coordinates of the minimum or maximum point.

12) $f(x) = 4 + 4x + x^2$

12) _____

Find the x-intercepts (if any) for the graph of the quadratic function.

13) $f(x) = 4 + 4x + x^2$

13) _____

Find the y-intercept for the graph of the quadratic function.

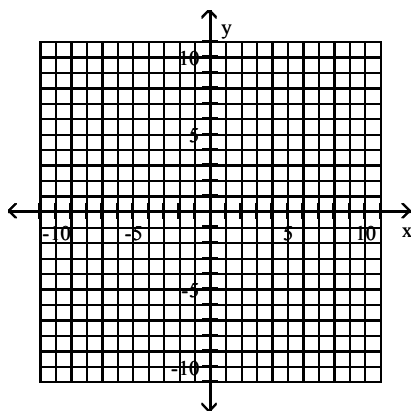
14) $f(x) = 4 + 4x + x^2$

14) _____

Using your answer from above, sketch the graph of the quadratic function.

15) $f(x) = 4 + 4x + x^2$

15) _____



Find the x-intercepts of the polynomial function. State whether the graph crosses the x-axis, or touches the x-axis and turns around, at each intercept.

16) $f(x) = -x^3(x + 4)^2(x - 9)$

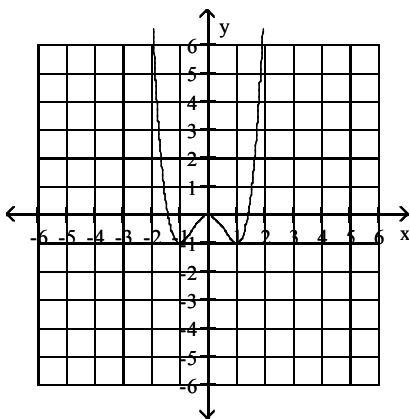
16) _____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

Determine whether the graph of the polynomial has y-axis symmetry, origin symmetry, or neither.

17)

17) _____



A) y-axis symmetry

B) origin symmetry

C) neither

Identify the leading coefficient and determine the end behavior of the polynomial function.

18) $f(x) = -3x^4 + 4x^3 - 5x^2 - 4x - 1$

18) _____

A) falls to the left and falls to the right

B) falls to the left and rises to the right

C) rises to the left and rises to the right

D) rises to the left and falls to the right

19) $f(x) = x^3 + 4x^2 + 2x - 4$

19) _____

A) falls to the left and falls to the right

B) falls to the left and rises to the right

C) rises to the left and rises to the right

D) rises to the left and falls to the right

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Find the zeros for the polynomial function and give the multiplicity for each zero. State whether the graph crosses the x-axis or touches the x-axis and turns around, at each zero.

20) $f(x) = 4(x - 2)(x + 1)^4$

20) _____

Use the Intermediate Value Theorem to determine whether the polynomial function has a real zero between the given integers.

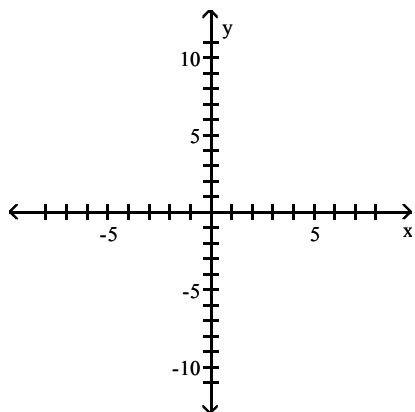
21) $f(x) = 9x^5 + 10x^3 - 5x^2 + 7$; between -1 and 0

21) _____

Graph the polynomial function.

22) $f(x) = 2x(x + 1)^2$

22) _____



Solve the problem.

- 23) The cost in millions of dollars for a company to manufacture x thousand automobiles is given by the function $C(x) = 3x^2 - 18x + 63$. Find the number of automobiles that must be produced to minimize the cost.

23) _____

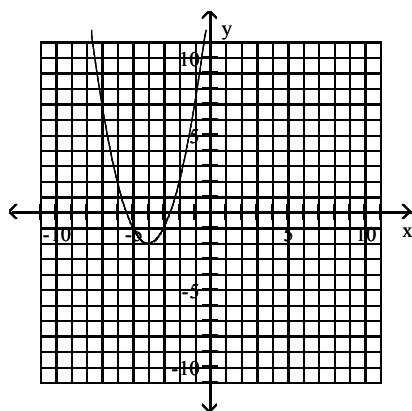
- 24) A developer wants to enclose a rectangular grassy lot that borders a city street for parking. If the developer has 236 feet of fencing and does not fence the side along the street, what is the largest area that can be enclosed?

24) _____

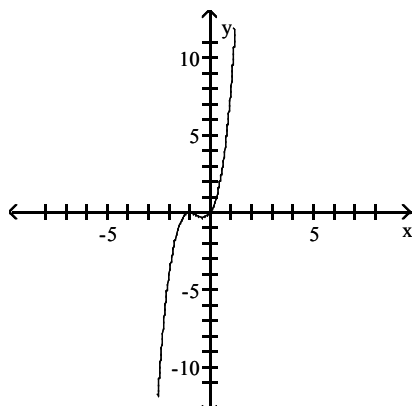
Answer Key

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- 1) $y = -2x + 4$
- 2) $y = -6x + 17$
- 3) B
- 4) A
- 5) Domain: $(-\infty, \infty)$
Range: $[-6, \infty)$
- 6) $(4, -4)$
- 7) $x = -3$
- 8) $[5, \infty)$
- 9) $(0, 0)$ and $(-4, 0)$
- 10) $(0, 0)$
- 11) $(-5, -17)$
- 12) minimum; $(-1, -10)$
- 13) $(-3, 0)$ and $(-2, 0)$
- 14) $(0, 8)$
- 15)



- 16) 0, crosses the x-axis;
-4, touches the x-axis and turns around;
9, crosses the x-axis
- 17) A
- 18) A
- 19) B
- 20) 2, multiplicity 1, crosses x-axis; -1, multiplicity 4, touches x-axis and turns around
- 21) $f(-1) = -17$ and $f(0) = 7$; yes
- 22)



- 23) 3 thousand automobiles

Answer Key

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24) 6962 ft²